Stage 1 Hard Drill — With Solutions

Chapter 3.3 — Selection Bias, Adverse Selection & Information Economics

10 new competition-grade questions, full-chapter coverage, full worked solutions. Target: under 90 seconds each once the shortcut is spotted.

Q1 — Conditioning on Execution, the Other Side

True value $V \sim \text{Unif}(0, 200)$. A market maker rests an ask at \$160. An informed trader buys *iff* $V > 160$. Find $\mathbb{E}[V \mid \text{buy order fills}]$ and the maker's expected giveaway per fill.

Solution. Conditional on a fill, $V \mid V > 160$ is uniform on $(160, 200)$:

$$\mathbb{E}[V \mid \text{fill}] = \frac{160 + 200}{2} = \boxed{\$180}$$

$$\text{giveaway} = \mathbb{E}[V \mid \text{fill}] - \text{ask} = 180 - 160 = \boxed{\$20}$$

⋆ *Intuition:* the ask only ever gets hit by a buyer who knows V exceeds it — truncating a uniform always moves the conditional mean to the midpoint of the *surviving* range, never the midpoint of the whole range. This mirrors Example 1's bid-side logic exactly, flipped to the ask.

Q2 — Capping the Overpayment on a Resting Bid

$V \sim \text{Unif}(0, 300)$. A resting bid at b only fills when an informed seller has $V < b$. Find b such that the average overpayment on a fill, $b - \mathbb{E}[V \mid \text{fill}]$, is capped at \$12.

Solution. Conditional on a fill, $V \mid V < b \sim \text{Unif}(0, b)$, so $\mathbb{E}[V \mid \text{fill}] = b/2$.

$$\text{overpayment} = b - \frac{b}{2} = \frac{b}{2} = 12$$

$$b = \boxed{\$24}$$

▶ *Speed trick:* for $V \sim \text{Unif}(0, M)$ truncated at a bid b, the overpayment is *always exactly* $b/2$ regardless of M — only the truncation point matters, since $\mathbb{E}[V \mid V < b] = b/2$ for any uniform starting at 0.

Q3 — Baseline Update, No Competition Yet

A jar's true value V is \$50 or \$150, equally likely. Each bidder's signal is correct w.p. $a = 0.75$ (equal to the other value w.p. 0.25). Bidder 1 sees $S_1 = 150$. Find $\mathbb{P}(V{=}150 \mid S_1{=}150)$ and $\mathbb{E}[V \mid S_1{=}150]$ — no competition yet.

Solution. Bayes:

$$\mathbb{P}(V{=}150 \mid S_1{=}150) = \frac{0.75(0.5)}{0.75(0.5) + 0.25(0.5)} = \boxed{\tfrac{3}{4}}$$

$$\mathbb{E}[V \mid S_1{=}150] = 150\left(\tfrac{3}{4}\right) + 50\left(\tfrac{1}{4}\right) = \boxed{\$125}$$

⋆ *Intuition:* identical machinery to Example 6, new numbers — this is *pure* signal-updating, with no auction dynamics folded in yet. Q4 adds the competitor.

Q4 — The Winner's Curse, Fully Derived

Same setup, 2 bidders, signals conditionally independent given V, accuracy $a = 0.75$. Bidder 1 wins (bids = signal, ties coin-flipped) iff $S_1 \geq S_2$. Given $S_1 = 150$, find $\mathbb{E}[V \mid S_1{=}150, S_2{=}50]$, $\mathbb{E}[V \mid S_1{=}150, S_2{=}150]$, $\mathbb{P}(\text{win} \mid S_1{=}150)$, and $\mathbb{E}[V \mid S_1{=}150, \text{win}]$.

Solution. Signals disagree ⇒ net cancel, back to the \$100 prior midpoint:

$$\mathbb{E}[V \mid S_1{=}150, S_2{=}50] = \boxed{\$100}$$

Signals agree: $\mathbb{P}(V{=}150 \mid S_1{=}S_2{=}150) = \dfrac{0.75^2}{0.75^2 + 0.25^2} = \dfrac{9}{10}$, so

$$\mathbb{E}[V \mid S_1{=}150, S_2{=}150] = 150\left(\tfrac{9}{10}\right) + 50\left(\tfrac{1}{10}\right) = \boxed{\$140}$$

$\mathbb{P}(S_2{=}50 \mid S_1{=}150) = 2a(1-a) = \frac{3}{8}$, so $\mathbb{P}(S_2{=}150 \mid S_1{=}150) = \frac{5}{8}$:

$$\mathbb{P}(\text{win} \mid S_1{=}150) = \tfrac{3}{8}(1) + \tfrac{5}{8}\left(\tfrac{1}{2}\right) = \boxed{\tfrac{11}{16}}$$

$$\mathbb{E}[V \mid S_1{=}150, \text{win}] = \frac{\frac{3}{8}(100) + \frac{5}{8}\cdot\frac{1}{2}\cdot 140}{11/16} = \boxed{\tfrac{1300}{11} \approx \$118.18}$$

▶ *Full-circle intuition:* three numbers, same signal: naive bid = \$150, signal-adjusted = \$125 (Q3), win-adjusted = \$118.18 (here). Winning means your rival more often had the *losing* signal — or tied you — both mildly bearish facts about V that a naive bidder never accounts for.

Q5 — The Boundary Case: an Uninformative Signal

Repeat Q4's full computation at $a = 0.5$ (a coin-flip signal, carrying zero information). Find $\mathbb{E}[V \mid S_1{=}150, \text{win}]$ and explain what happens to the curse.

Solution. At $a = 0.5$: $\mathbb{E}[V \mid S_1{=}150, S_2{=}50] = 100$ (unchanged, still midpoint); agreeing branch gives $\mathbb{P}(V{=}150 \mid \text{agree}) = \frac{0.25}{0.25+0.25} = \frac{1}{2} \Rightarrow \mathbb{E} = 100$ too.

$$\mathbb{P}(\text{win} \mid S_1{=}150) = \tfrac{1}{2}(1) + \tfrac{1}{2}\left(\tfrac{1}{2}\right) = \tfrac{3}{4}$$

$$\mathbb{E}[V \mid S_1{=}150, \text{win}] = \frac{\frac{1}{2}(100) + \frac{1}{2}\cdot\frac{1}{2}\cdot 100}{3/4} = \boxed{\$100}$$

The curse vanishes entirely — win-adjusted equals the raw \$100 prior.

⋆ *Intuition:* a coin-flip signal carries no information to begin with, so conditioning on anything built from it — including winning — cannot move your estimate. The winner's curse is not a generic auction tax; it is a correction for information that the signal (and hence the win event) actually contains. Zero signal quality ⇒ zero curse.

Q6 — The Closed-Form Spread, Direct Substitution

$V \in \{80, 120\}$ equally likely. A fraction $\alpha = 0.25$ of order flow is informed. Using the closed-form formula, find the ask, bid, and spread.

Solution. FV = 100, $\frac{H-L}{2} = 20$:

$$\text{ask} = 100 + 0.25(20) = \boxed{\$105}, \qquad \text{bid} = 100 - 0.25(20) = \boxed{\$95}$$

$$\text{spread} = \alpha(H - L) = 0.25(40) = \boxed{\$10}$$

▶ *Speed trick:* one substitution into ask $= \text{FV} + \alpha\frac{H-L}{2}$, spread $= \alpha(H - L)$ — no Bayes required when the prior is symmetric 50/50.

Q7 — Reverse-Engineering α from an Observed Ask

Same \$80/\$120 contract, symmetric prior. A competitor's ask is observed at \$114. Back out their assumed α.

Solution.

$$\text{ask} = 100 + \alpha(20) = 114$$

$$\alpha = \frac{14}{20} = \boxed{0.7}$$

⋆ *Intuition:* exactly like Example 15's spread-based read, but using only one side of the quote — the ask alone is enough to recover α once you know FV and $\frac{H-L}{2}$, since each is linear in α with a known slope.

Q8 — Full Bayes, Asymmetric Prior, No Shortcut

$V \in \{80, 120\}$, but now $\mathbb{P}(V{=}120) = 0.6$ (asymmetric prior), $\alpha = 0.5$. Find the ask, bid, and spread — the linear shortcut does not apply.

Solution. $\mathbb{P}(\text{buy} \mid 120) = \frac{1+\alpha}{2} = 0.75$, $\mathbb{P}(\text{buy} \mid 80) = 0.25$.

$$\mathbb{P}(\text{buy}) = 0.6(0.75) + 0.4(0.25) = 0.55$$

$$\mathbb{P}(120 \mid \text{buy}) = \frac{0.45}{0.55} = \frac{9}{11}$$

$$\text{ask} = 120\left(\tfrac{9}{11}\right) + 80\left(\tfrac{2}{11}\right) = \boxed{\tfrac{1240}{11} \approx \$112.73}$$

Symmetrically, $\mathbb{P}(\text{sell}) = 0.6(0.25) + 0.4(0.75) = 0.45$, $\mathbb{P}(80 \mid \text{sell}) = \frac{0.3}{0.45} = \frac{2}{3}$:

$$\text{bid} = 80\left(\tfrac{2}{3}\right) + 120\left(\tfrac{1}{3}\right) = \boxed{\tfrac{280}{3} \approx \$93.33}$$

$$\text{spread} = \tfrac{1240}{11} - \tfrac{280}{3} = \tfrac{640}{33} \approx \boxed{\$19.39}$$

▶ *Intuition:* exactly Example 16's lesson, new numbers — once the prior leaves 50/50, ask and bid no longer sit symmetrically around the naive fair value $0.6(120)+0.4(80) = \$104$; here the ask is pulled *closer* to FV than the bid is, because the prior already leans toward H.

Q9 — Stacking Information Risk on Inventory Risk

$V \in \{80, 120\}$ (FV $= 100$, $\sigma^2 = 400$), $\alpha = 0.5$ (info half-spread \$10). A maker holds $q = 3$, $\gamma = 0.02$. Find the final two-sided quote.

Solution. Reservation price:

$$r = \text{FV} - q\gamma\sigma^2 = 100 - (3)(0.02)(400) = \boxed{\$76}$$

Info half-spread ($\alpha \cdot \frac{H-L}{2} = 0.5(20) = 10$), applied around r instead of FV:

$$\text{bid} = 76 - 10 = \boxed{\$66}, \qquad \text{ask} = 76 + 10 = \boxed{\$86}$$

⋆ *Intuition:* same two-step recipe as Example 19 — inventory risk relocates the *center* (\$100 → \$76), information risk sets the *width* (±\$10) around wherever that center ends up. The two skews never interact algebraically; compute each in isolation, then add.

Q10 — Full Capstone: Asymmetric Prior, Inventory, Both Sides Unequal

$V \in \{60, 140\}$, $\mathbb{P}(V{=}140) = 0.4$, $\alpha = 0.4$. Maker is short $q = -6$, $\gamma = 0.004$. Find the final bid/ask.

Solution. FV $= 0.4(140) + 0.6(60) = \$92$, Var $= 0.4(0.6)(80)^2 = 1536$. Full Bayes: $\mathbb{P}(\text{buy} \mid 140) = 0.7$, $\mathbb{P}(\text{buy} \mid 60) = 0.3 \Rightarrow \mathbb{P}(\text{buy}) = 0.46$,

$$\mathbb{P}(140 \mid \text{buy}) = \tfrac{0.28}{0.46} = \tfrac{14}{23} \Rightarrow \text{GM ask} = \boxed{\tfrac{2500}{23} \approx \$108.70}$$

$\mathbb{P}(\text{sell} \mid 140) = 0.3$, $\mathbb{P}(\text{sell} \mid 60) = 0.7 \Rightarrow \mathbb{P}(\text{sell}) = 0.54$,

$$\mathbb{P}(60 \mid \text{sell}) = \tfrac{0.42}{0.54} = \tfrac{7}{9} \Rightarrow \text{GM bid} = \boxed{\tfrac{700}{9} \approx \$77.78}$$

Info half-spreads are *unequal* on each side: ask-side $= \frac{2500}{23} - 92 \approx 16.70$, bid-side $= 92 - \frac{700}{9} \approx 14.22$. Reservation price:

$$r = 92 - (-6)(0.004)(1536) = 92 + 36.864 = \boxed{\$128.864}$$

Apply each side's *own* info half-spread around r:

$$\text{ask} = 128.864 + 16.70 \approx \boxed{\$145.56}$$

$$\text{bid} = 128.864 - 14.22 \approx \boxed{\$114.64}$$

▶ *Full-circle intuition:* the entire chapter in one quote — an asymmetric prior breaks Example 14's clean symmetric spread (Section C), so the ask-side and bid-side info widths must be computed *separately* (here \$16.70 vs. \$14.22, not a shared ±\$15.46), and only then stacked onto an inventory-shifted center (Ch. 3.2). A short position in a contract already skewed bullish by its prior pushes the reservation price dramatically above FV — exactly the layered, asymmetric risk-stacking a Stage 1 "hard" question is designed to test under time pressure.

Drill order for review: re-derive Q4 (the full winner's-curse Bayes chain, tie-break included) and Q5 (explain in one sentence why *zero signal quality kills the curse) closed-book, then Q10 (asymmetric GM stacked on inventory) end-to-end. Together they test whether conditioning-on-execution, the winner's curse, and Glosten–Milgrom run as one connected Bayesian skill, not three separate formulas.*